

CO3_AT2

18. Case Study: Student Record Retrieval System (Binary Search)

A university database stores student details in sorted order and requires quick search operations. Analyze how Binary Search improves retrieval efficiency. Identify requirements such as sorted data and accuracy. Apply divide and conquer principles. Analyze time complexity. Design a solution and justify its effectiveness.

SOURCE CODE:

Student Record Retrieval System using Binary Search

```
student_ids = [101, 105, 110, 115, 120, 125, 130]
student_names = ["Deekshu", "Arun", "Bala", "Charan", "Divya", "Esha",
"Farhan"]
```

```
key = int(input("Enter Student ID to Search: "))
```

```
low = 0
```

```
high = len(student_ids) - 1
```

```
found = False
```

```
while low <= high:
```

```
    mid = (low + high) // 2
```

```
    if student_ids[mid] == key:
```

```
        print("\nStudent Record Found")
```

```
        print("-----")
```

```
        print("Student ID  :", student_ids[mid])
```

```
        print("Student Name :", student_names[mid])
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```
        found = True
```

```
        break
```

```
    elif key < student_ids[mid]:
```

```
        high = mid - 1
```

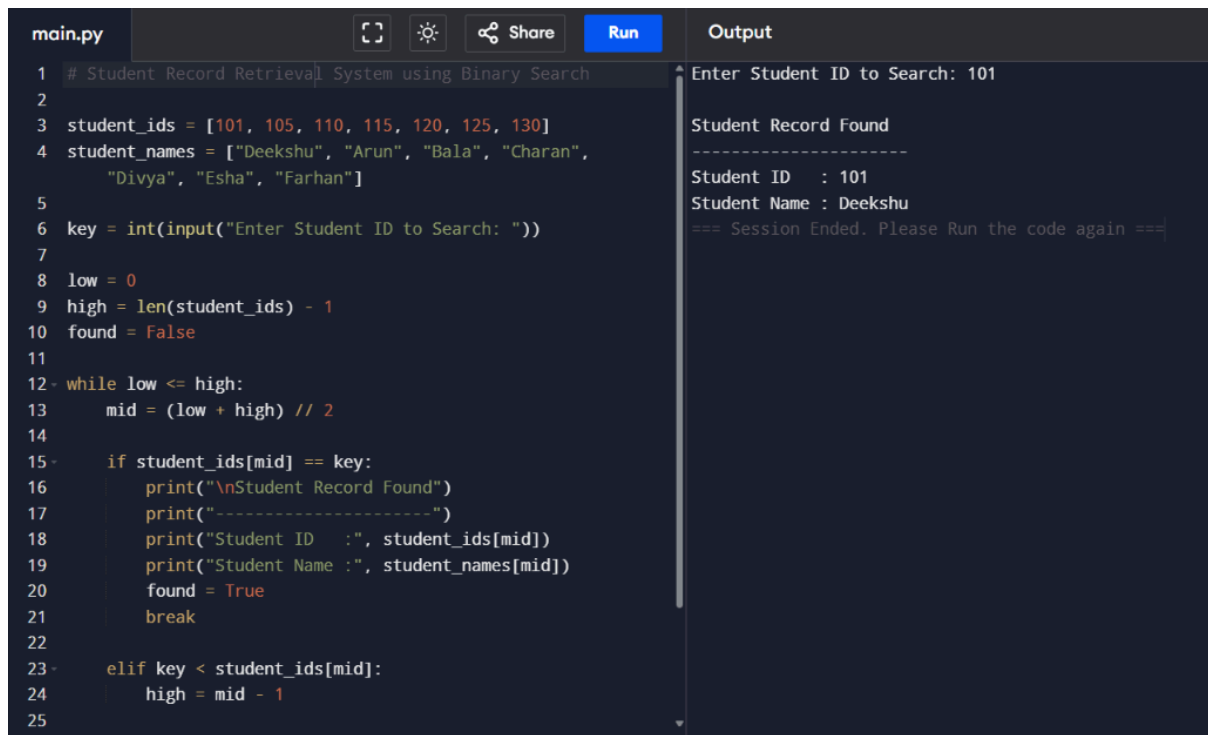
```
    else:
```

low = mid + 1

if not found:

print("\nStudent Record Not Found")

OUTPUT:



The screenshot displays a code editor with a file named 'main.py'. The code implements a binary search algorithm to find a student record based on their ID. The student IDs are [101, 105, 110, 115, 120, 125, 130] and the corresponding names are ["Deekshu", "Arun", "Bala", "Charan", "Divya", "Esha", "Farhan"]. The program prompts the user to enter a student ID. In the output window, the user has entered '101', and the program successfully finds the record for 'Deekshu'. The output also includes a session-ending message.

```
main.py  [Icons]  Run  Output
1 # Student Record Retrieval System using Binary Search
2
3 student_ids = [101, 105, 110, 115, 120, 125, 130]
4 student_names = ["Deekshu", "Arun", "Bala", "Charan",
5                  "Divya", "Esha", "Farhan"]
6 key = int(input("Enter Student ID to Search: "))
7
8 low = 0
9 high = len(student_ids) - 1
10 found = False
11
12 while low <= high:
13     mid = (low + high) // 2
14
15     if student_ids[mid] == key:
16         print("\nStudent Record Found")
17         print("-----")
18         print("Student ID   :", student_ids[mid])
19         print("Student Name :", student_names[mid])
20         found = True
21         break
22
23     elif key < student_ids[mid]:
24         high = mid - 1
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```

Enter Student ID to Search: 101

Student Record Found

Student ID : 101

Student Name : Deekshu

=== Session Ended. Please Run the code again ===

